1 Scientific user

Natural-language request

"Find HST and JWST observations near NGC 6543 and relevant literature."

2 AstroAKD agent

Domain-aware orchestration

resolve targets

extract constraints

AstroAKD agent

select archives

compose queries

3 MCP tool layer

Standardized tool interface

MCP server

discover

inspect

execute

normalize

4 Archive access layer

Astroquery-based access

Astroquery

converts requests into coordinates, units, and archive-specific calls

5 Astronomical archives & services

Distributed astronomical data and knowledge resources

SIMBAD

VizieR

ADS

VO resources

MAST

Gaia

HEASARC

NED

IRSA

6 Evidence, provenance, and outputs

Benchmark requests, MCP calls, and outputs retained for validation and repeatability.

candidate datasets



Relevant observations from multiple archives

metadata tables

RA	Dec	Instrument	ObsID
287.42	66.64	HST/WFC3	icxe01010
287.42	66.64	JWST/NIRCam	jw012340
...	...	...	...

Harmonized and queryable results

archive provenance

Query target: NGC 6543
Archives: MAST, ADS, ...
MCP calls: discover → inspect → execute
Provenance: dataset IDs, DOIs, citations
Timestamps: 2024-11-08T14:23:18Z

Sources, queries, and citations for all results

inspectable tool traces

```
[1/4] discover     ✓ 2.3s
[2/4] inspect      ✓ 1.1s
[3/4] execute      ✓ 4.8s
[4/4] normalize    ✓ 0.9s
Retrieved 12 datasets, 8 papers
```

Full trace of tool calls and responses